



MICROSOFT EMS COMMUNITY

WELCOME TO

Apple Device Administration Essentials

A Beginner's Guide/Best Practices Deploying Apple Devices

4th September 2026

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TODAY'S SPEAKER

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ING

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<https://intuneirl.com>

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FAST PACE UNCOMPROMISED FOCUS

A NOTE FROM YOUR SPEAKER:

As someone living with Parkinson's, my speech can sometimes accelerate. My presentation is structured for clarity to ensure you stay on track, regardless of my speed.



Thank you for your support and understanding.

Where we're going in 60 minutes

01

Foundations & Planning

Apple ecosystem & deployment models

Apple Business Manager, Managed Apple Accounts

Enrollment methods, supervision, MDM choice

Setup Assistant

02

Identity, Config & Content

Network & APNs prerequisites

Platform SSO and Microsoft Entra ID

Profiles, DDM, software updates

Apps & Books, networks, content caching

03

Security, Scaling & Wrap-up

FileVault, system security, attestation

Compliance & Conditional Access

Platform differences, migration, troubleshooting

Scaling, governance, takeaways

New: WWDC26 & what changes with OS 27

PART 01

Foundations & Planning

Apple ecosystem

Apple Business / School Manager

Managed Apple Accounts

Enrollment methods

Choosing an MDM

Setup Assistant

Intro to Apple platform deployment

Apple devices are designed to be managed as an integrated system — hardware, software, apps and services working together through six core building blocks.

01 · IDENTITY

Managed Apple Accounts

Org-owned Apple identity federated to Entra ID, separate from personal Apple IDs.

02 · ENROLLMENT

ABM + Automated Device Enrollment

Devices tied to your org at purchase; supervision and non-removable management.

03 · CONFIGURATION

Profiles & declarations

MDM payloads and DDM declarations that set policy without user interaction.

04 · CONTENT

Apps & Books

VPP licensing, custom apps, PKG/DMG delivery — silent, no Apple ID needed.

05 · SECURITY

Built-in protections

Secure Enclave, FileVault, Gatekeeper, SIP — MDM enforces policy on top.

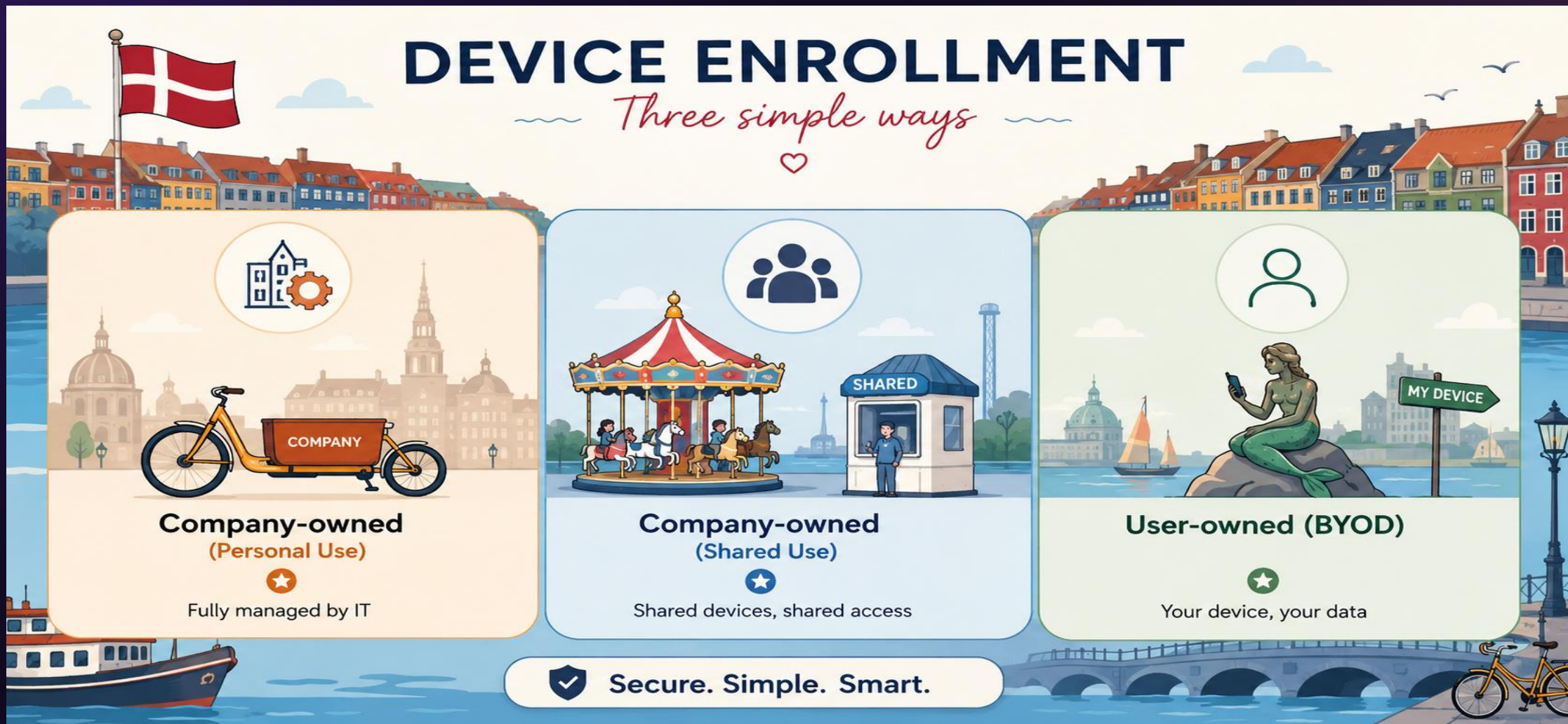
06 · SERVICES

APNs, caching, updates

The plumbing everything else rides on. Test it before day one.

Choosing a deployment model

Your deployment model determines the enrollment method and the level of IT control. Apple supports three primary models.



Device supervision

Supervision is a special mode that gives IT full management control. It can only be enabled through ADE or Apple Configurator — not after initial setup without a wipe.



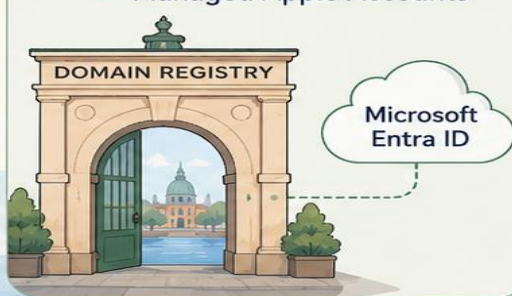
Apple Business Manager — setup & overview

ABM is your organization's free portal to Apple's deployment ecosystem — device enrollment, app licensing and identity. Start early; Apple's verification takes time.



APPLE BUSINESS MANAGER

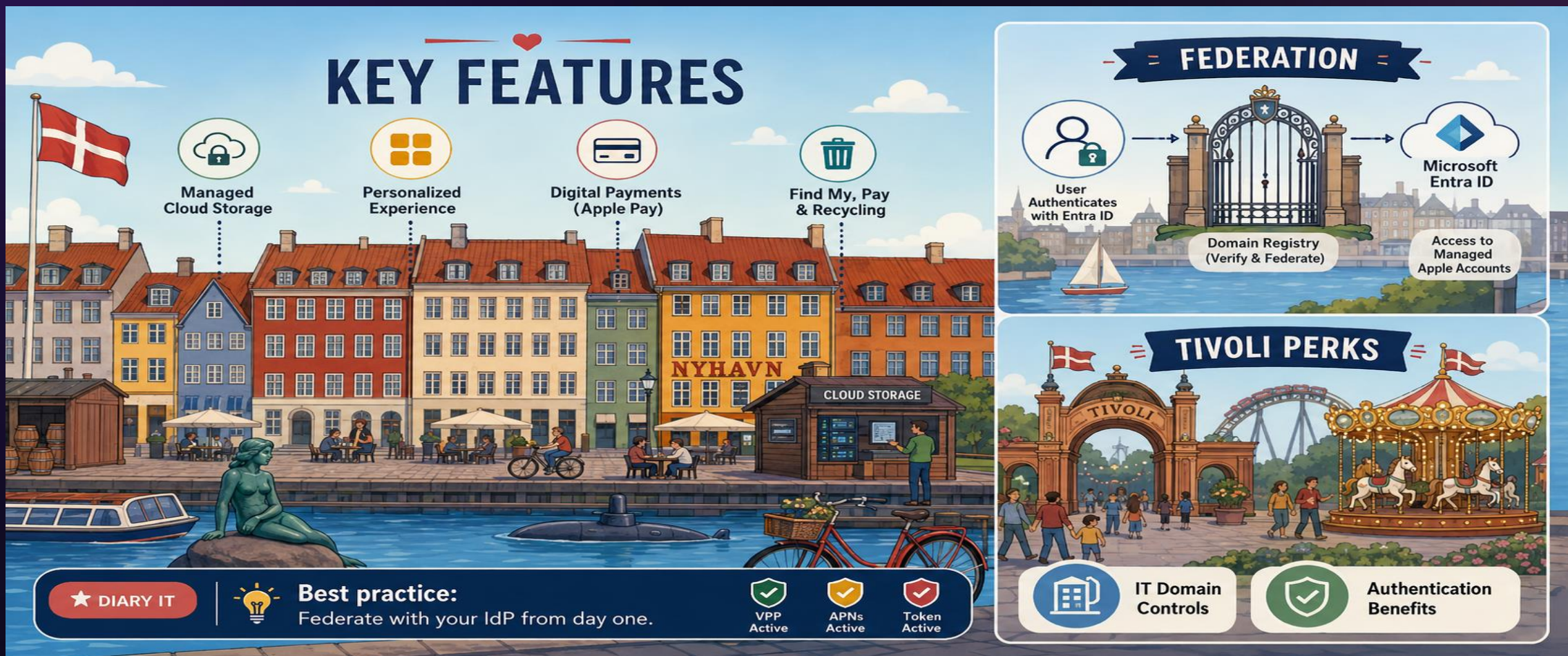
Set up in a few key steps

- 1. Enroll organization**
 - D-U-N-S number
 - Business contact
 - Dedicated Apple ID
- 2. Claim domains**
 - Verify DNS
 - Federate with Microsoft Entra ID
 - Managed Apple Accounts
- 3. Connect & configure**
 - Link Intune MDM
 - Add reseller IDs
 - Set Apps & Books / admin roles

**Reminder:** Renew ABM token, APNs certificate, and Apps & Books token every year. If one expires, enrollment can stop.

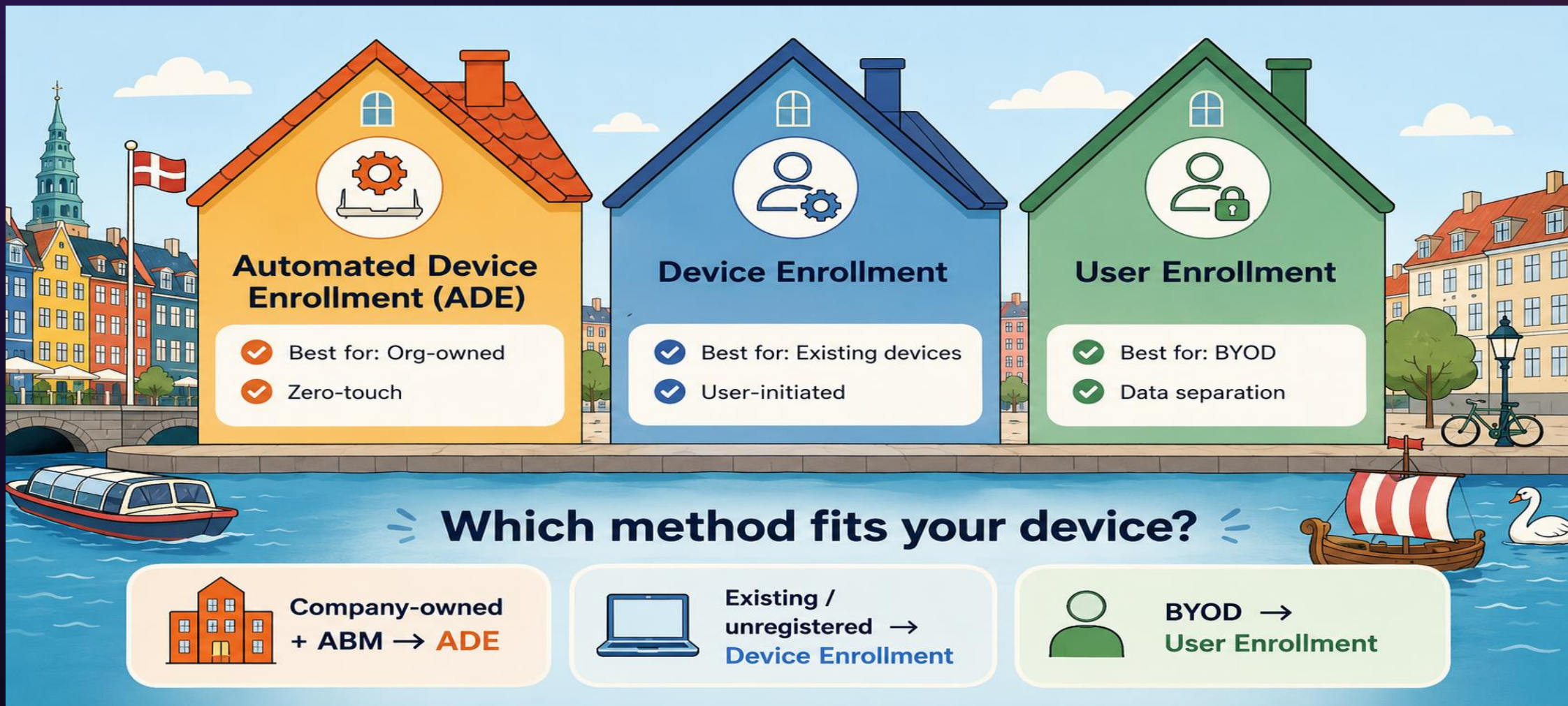
Managed Apple Accounts

Organizational Apple IDs created and owned by your company through ABM — separate from personal Apple IDs and fully under IT control.



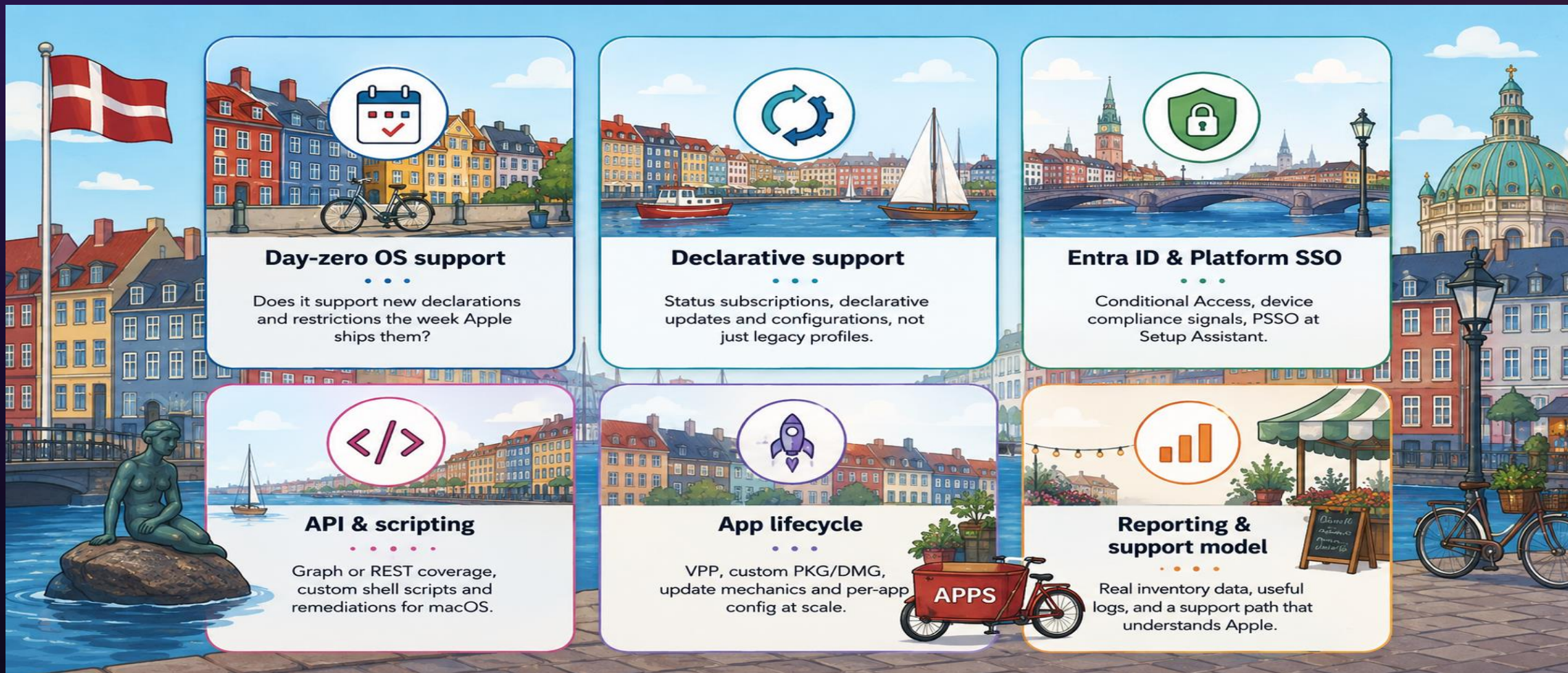
Enrollment methods compared

Three methods, each suited to a different ownership scenario. The right choice depends on who owns the device.



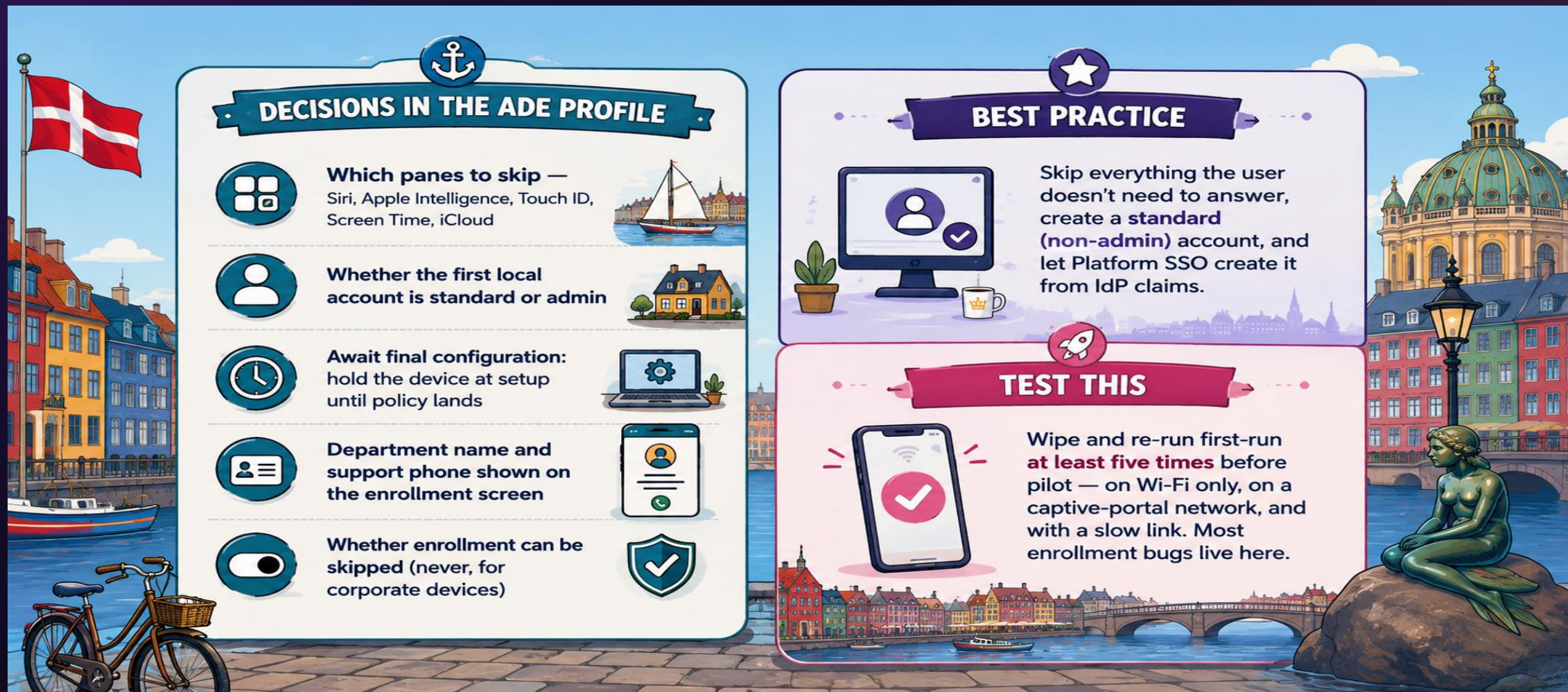
Choosing a device management service

Your MDM is the central platform for managing Apple devices. Evaluate candidates against these criteria before committing.



Setup Assistant — the first-run experience

Setup Assistant is what users see on first power-on. With MDM and ADE you control every screen they encounter.



DECISIONS IN THE ADE PROFILE

-  **Which panes to skip** — Siri, Apple Intelligence, Touch ID, Screen Time, iCloud
-  **Whether the first local account is standard or admin**
-  **Await final configuration:** hold the device at setup until policy lands
-  **Department name and support phone shown on the enrollment screen**
-  **Whether enrollment can be skipped** (never, for corporate devices)

BEST PRACTICE

Skip everything the user doesn't need to answer, create a **standard (non-admin)** account, and let Platform SSO create it from IdP claims.

TEST THIS

Wipe and re-run first-run **at least five times** before pilot — on Wi-Fi only, on a captive-portal network, and with a slow link. Most enrollment bugs live here.

PART 02

Identity, Configuration & Content

Network & APNs

Platform SSO

Entra ID integration

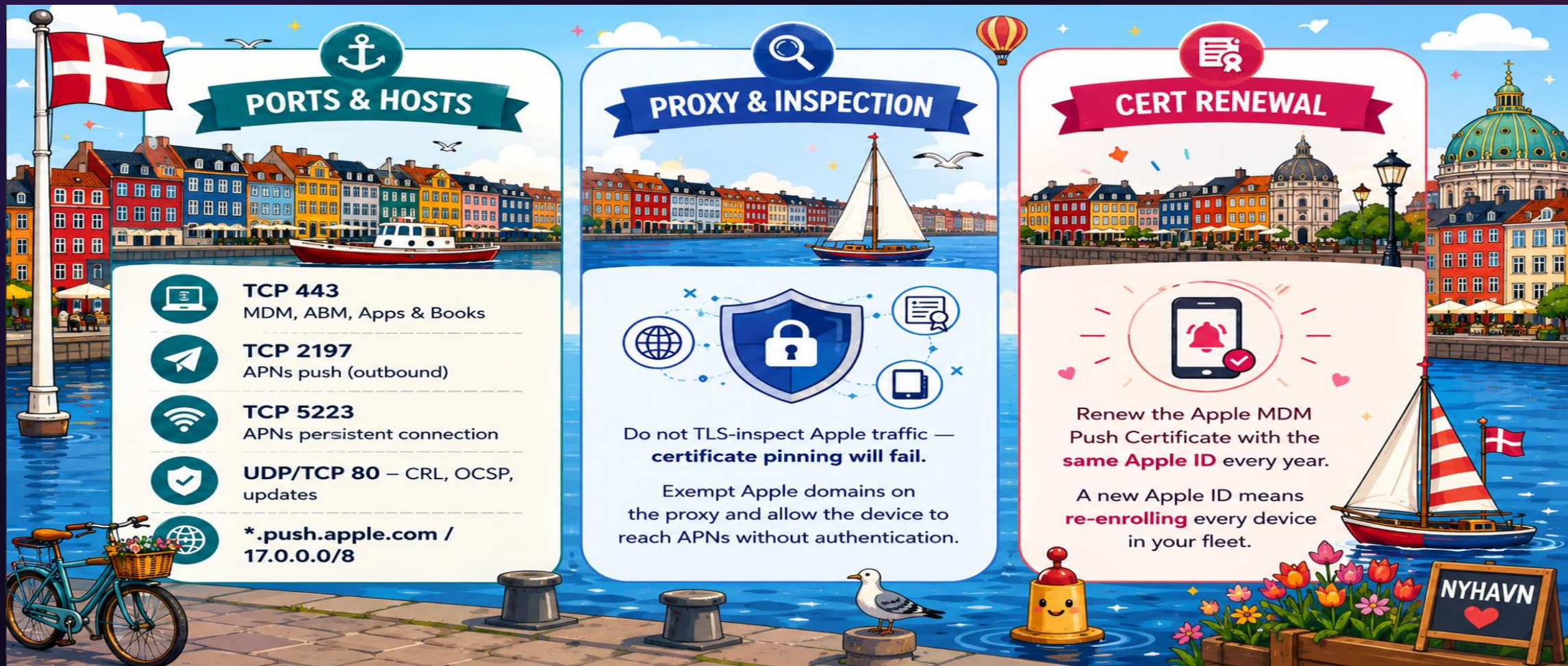
Profiles & DDM

Apps & updates

Networks & caching

Prepare your environment — network & APNs

Test APNs connectivity before enrollment. A silently blocked port 443 or 2197 breaks MDM with no clear error.



PORTS & HOSTS

- TCP 443**
MDM, ABM, Apps & Books
- TCP 2197**
APNs push (outbound)
- TCP 5223**
APNs persistent connection
- UDP/TCP 80** – CRL, OCSP, updates
- *.push.apple.com / 17.0.0.0/8**

PROXY & INSPECTION

Do not TLS-inspect Apple traffic — **certificate pinning will fail.**

Exempt Apple domains on the proxy and allow the device to reach APNs without authentication.

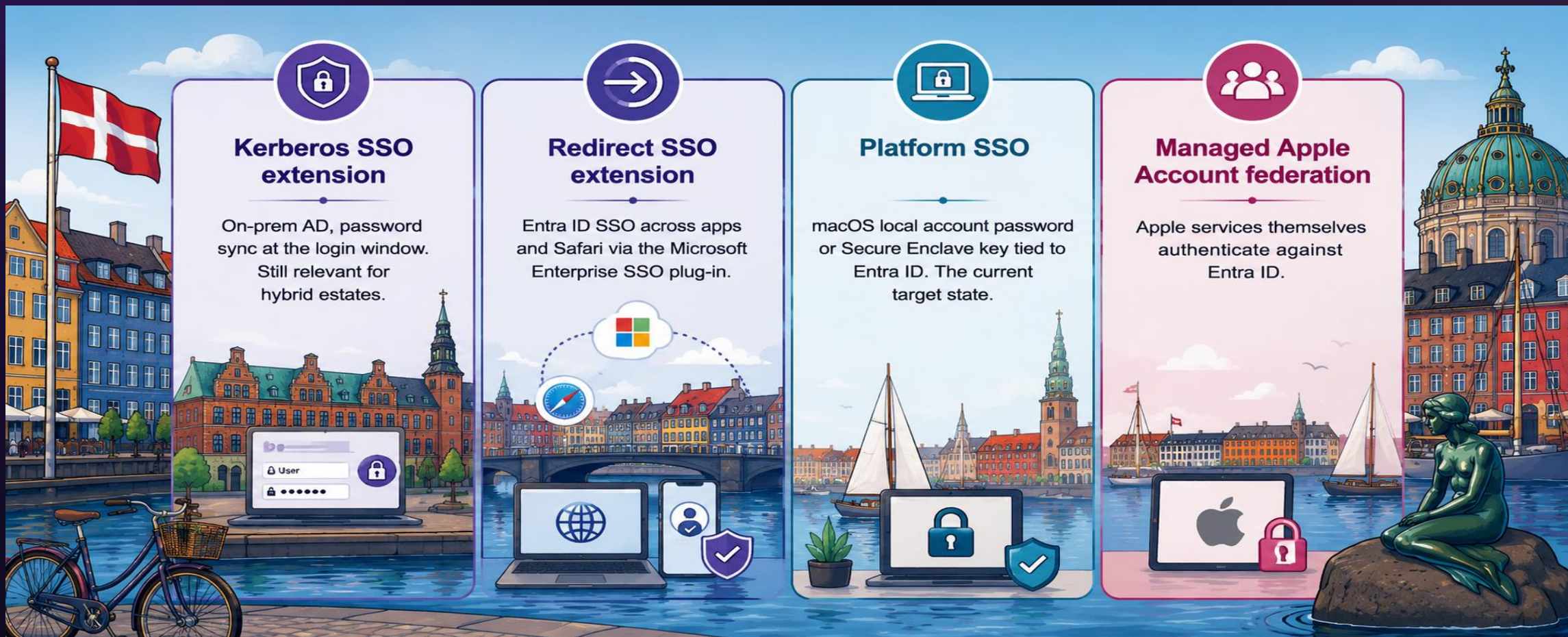
CERT RENEWAL

Renew the Apple MDM Push Certificate with the **same Apple ID** every year.

A new Apple ID means **re-enrolling** every device in your fleet.

Why identity integration matters

Apple devices support enterprise identity services for seamless, secure authentication — less friction for users, stronger compliance for IT.



DEMO

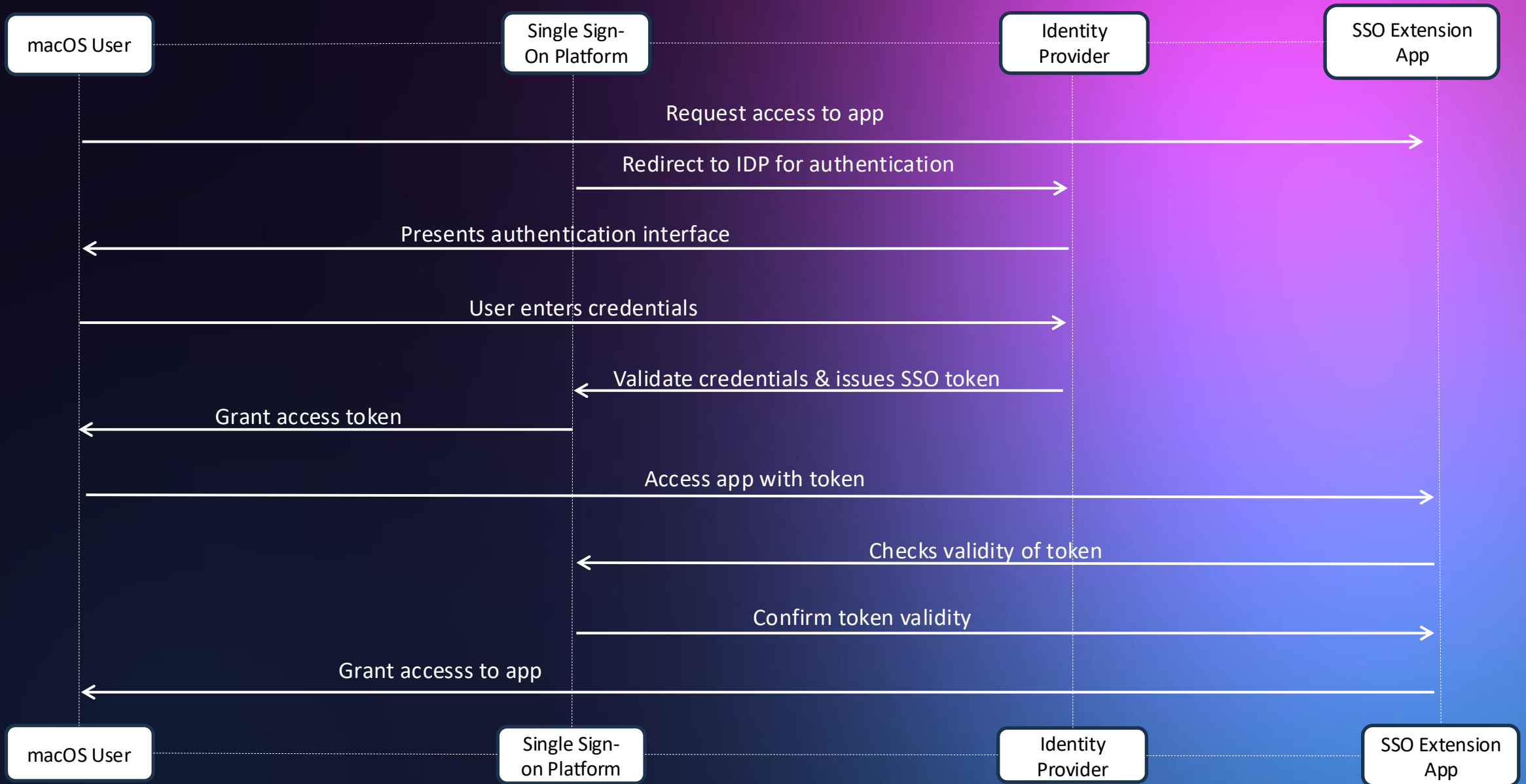
Platform SSO, apps and updates

PSSO registration & login window

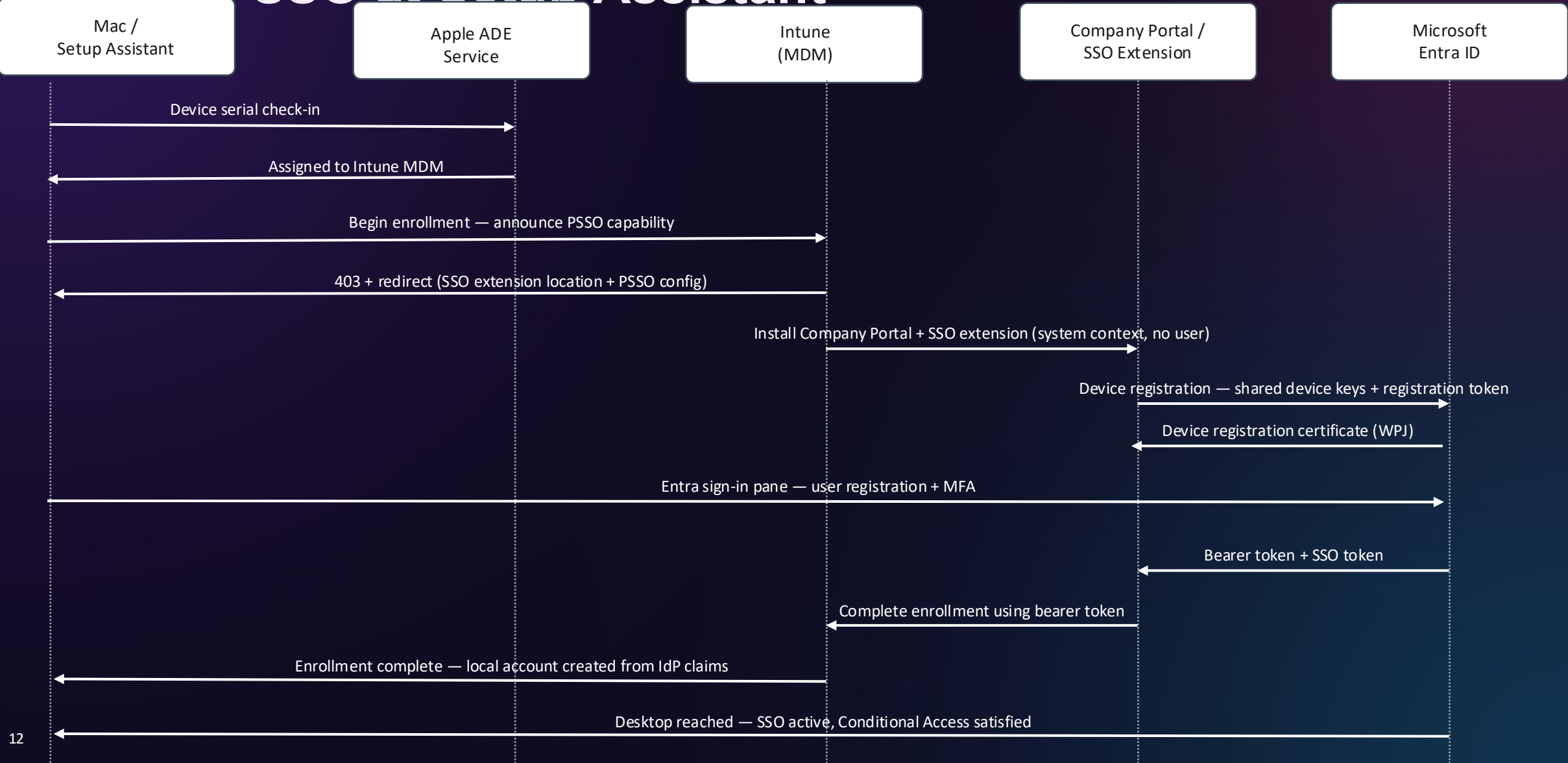
VPP app assignment

Declarative update deadline

Platform SSO for macOS (Old Method)

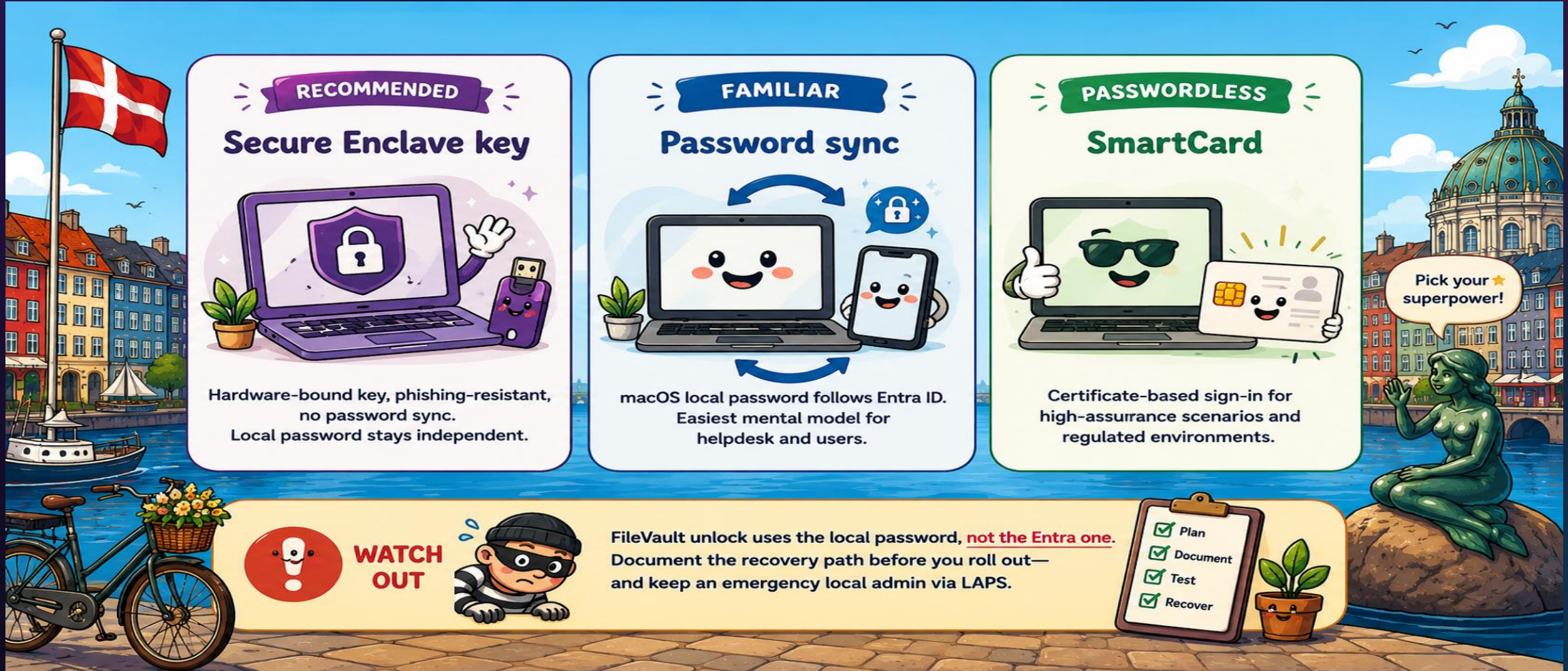


Platform SSO at Setup Assistant



Integrate with Microsoft Entra ID

Platform SSO offers three authentication methods. Choose per device ring, and decide up front how local passwords behave.



RECOMMENDED

Secure Enclave key



Hardware-bound key, phishing-resistant,
no password sync.
Local password stays independent.

FAMILIAR

Password sync



macOS local password follows Entra ID.
Easiest mental model for
helpdesk and users.

PASSWORDLESS

SmartCard



Certificate-based sign-in for
high-assurance scenarios and
regulated environments.

WATCH OUT



FileVault unlock uses the local password, **not the Entra one.**
Document the recovery path before you roll out—
and keep an emergency local admin via LAPS.



- ✓ Plan
- ✓ Document
- ✓ Test
- ✓ Recover

Pick your ★ superpower!

Declarative Device Management

The device becomes an active participant: it applies declarations autonomously and reports status changes to the MDM instead of waiting to be asked.

BUILDING BLOCKS

- Configurations** — what to apply (passcode, updates, apps)
- Assets** — reusable data referenced by configurations
- Activations** — atomic sets: all apply, or none
- Predicates** — conditions the device evaluates itself
- Status subscriptions** — the device pushes state changes

WHY IT MATTERS

Faster, more reliable policy application; real status reporting instead of inventory guesswork; managed software updates that actually enforce a deadline.

IN PRACTICE

Legacy profiles and declarations coexist. Where both exist for a setting, DDM wins — so migrate deliberately and don't double-configure.

Think LEGO: small bricks, big impact!

Configuration profiles & baselines

Payloads pushed silently via MDM, scoped at device or user level — no user interaction required.

01 SETTINGS CATALOG

Start here

Prefer the settings catalog over custom .mobileconfig — it reports properly and survives OS changes.



02 CUSTOM PROFILES

When you must

Signed .mobileconfig for payloads the catalog lacks. Version them in Git and document every key.



03 SCOPING

Device vs user

Security baselines at device scope; preferences and per-app config at user scope.



04 BASELINES

Use a known standard

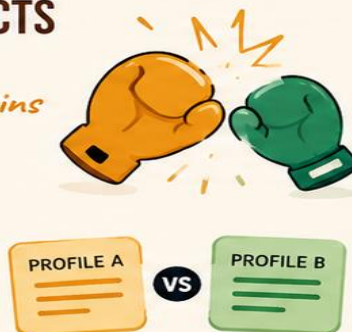
CIS Level 1 or the macOS Security Compliance Project as your reference, then justify each deviation.



05 CONFLICTS

Most restrictive wins

Two profiles setting the same key is a support ticket waiting to happen. One owner per setting.



06 SCRIPTS

Last resort on macOS

Shell scripts fill gaps but have no state model. Make them idempotent and log to a known path.



Manage software updates

Apple ships a major OS every September and security fixes continuously. Your job is to control the pace without blocking patches.

01

DECLARATIVE UPDATES



Target a specific OS version with a **hard enforcement** deadline. The device schedules, downloads and nags on its own — and reports progress back.



02

DEFERRALS



Defer major upgrades up to **90 days** while allowing minor and rapid security responses immediately. Never defer everything.

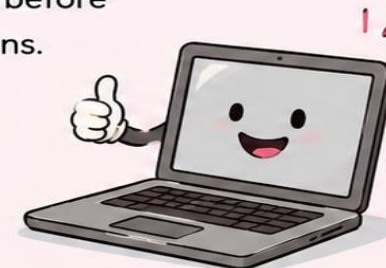


03

RINGS



IT ring → early adopters → broad. Test business-critical apps and **security tooling** in each ring before the next opens.



macOS REALITY

Updates need power, free space and a bootstrap token escrowed to the MDM. Without the **bootstrap token**, silent updates on Apple silicon simply won't apply.



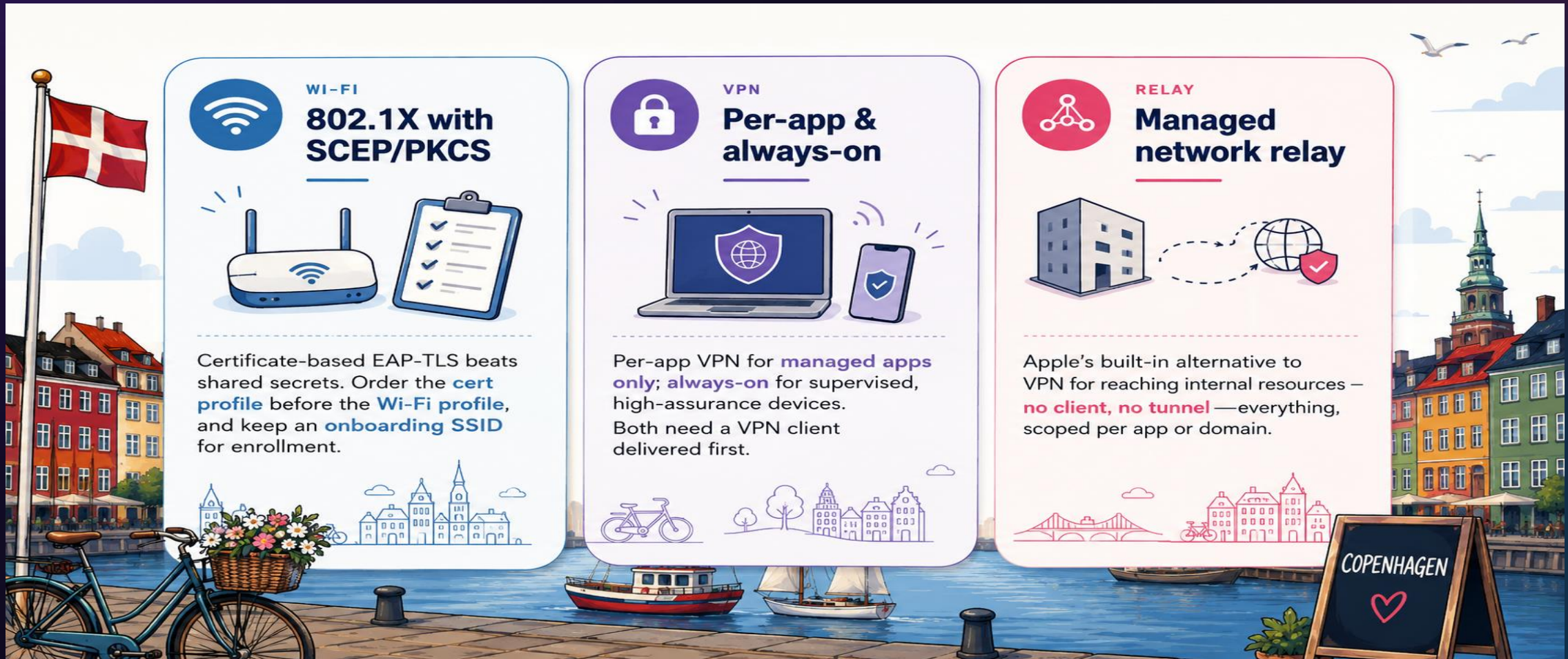
Distribute content — apps & books

Apps & Books (formerly VPP) is the central hub for licensing and distributing apps at scale — silently, with no Apple ID required on supervised devices.



Networks — Wi-Fi, VPN & relays

Push network configuration before the user needs it. On day one the device should join the corporate network without anyone typing a password.



WI-FI
**802.1X with
SCEP/PKCS**



Certificate-based EAP-TLS beats shared secrets. Order the **cert profile** before the **Wi-Fi profile**, and keep an **onboarding SSID** for enrollment.

VPN
**Per-app &
always-on**



Per-app VPN for **managed apps only**; **always-on** for supervised, high-assurance devices. Both need a VPN client delivered first.

RELAY
**Managed
network relay**



Apple's built-in alternative to VPN for reaching internal resources — **no client, no tunnel** — everything, scoped per app or domain.

COPENHAGEN



Content caching — speed up deployments

A built-in macOS service that caches Apple content locally, so the OS update or app downloads once per site instead of once per device.



PART 03

Security & Scaling

Securing devices

Compliance & Conditional Access

Platform differences

Migration & troubleshooting

Scaling & governance

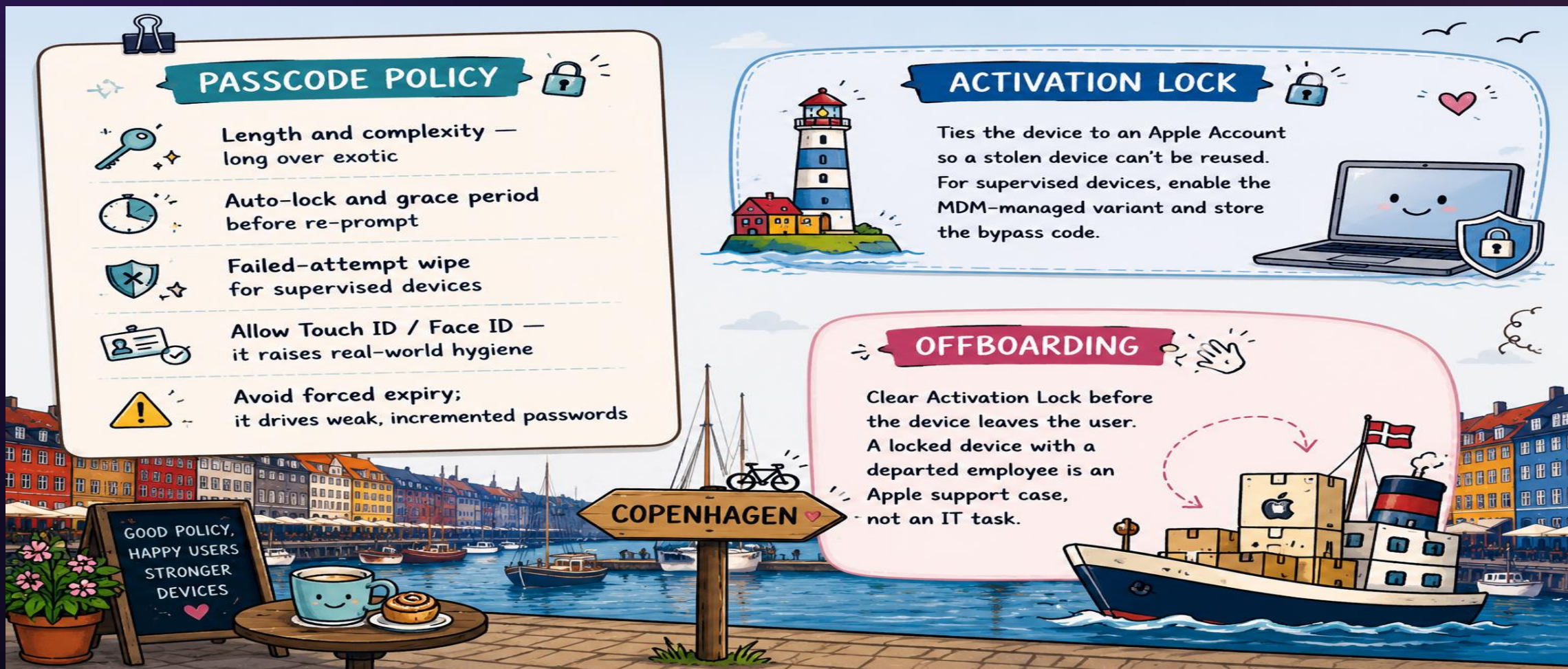
Ensure device security — overview

Apple devices ship with deep, layered security built in. MDM's role is to enforce organizational policy on top of that foundation.



Passcode policies & Activation Lock

Two everyday controls with outsized operational impact — one on users, one on your asset lifecycle.



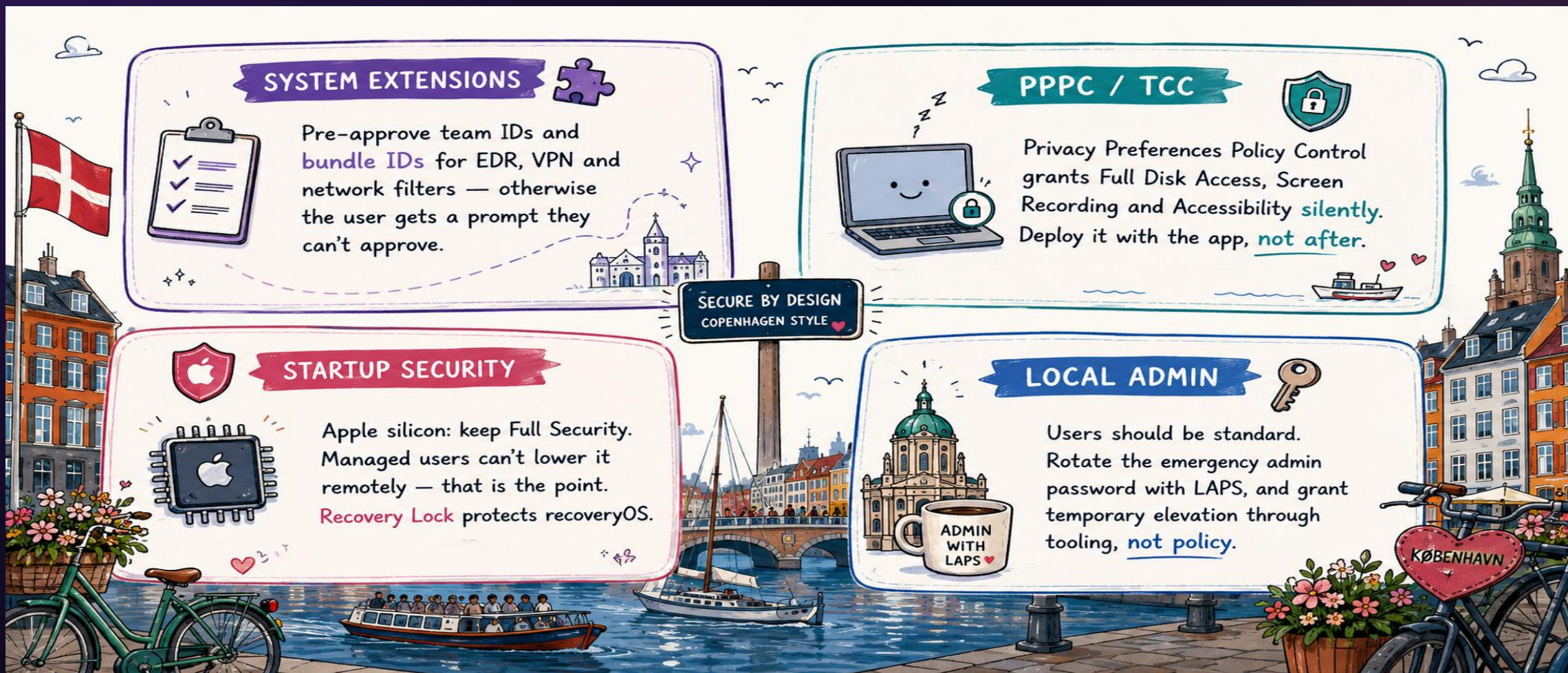
FileVault & disk encryption

FileVault is macOS full-disk encryption — essential for protecting data on lost or stolen machines.



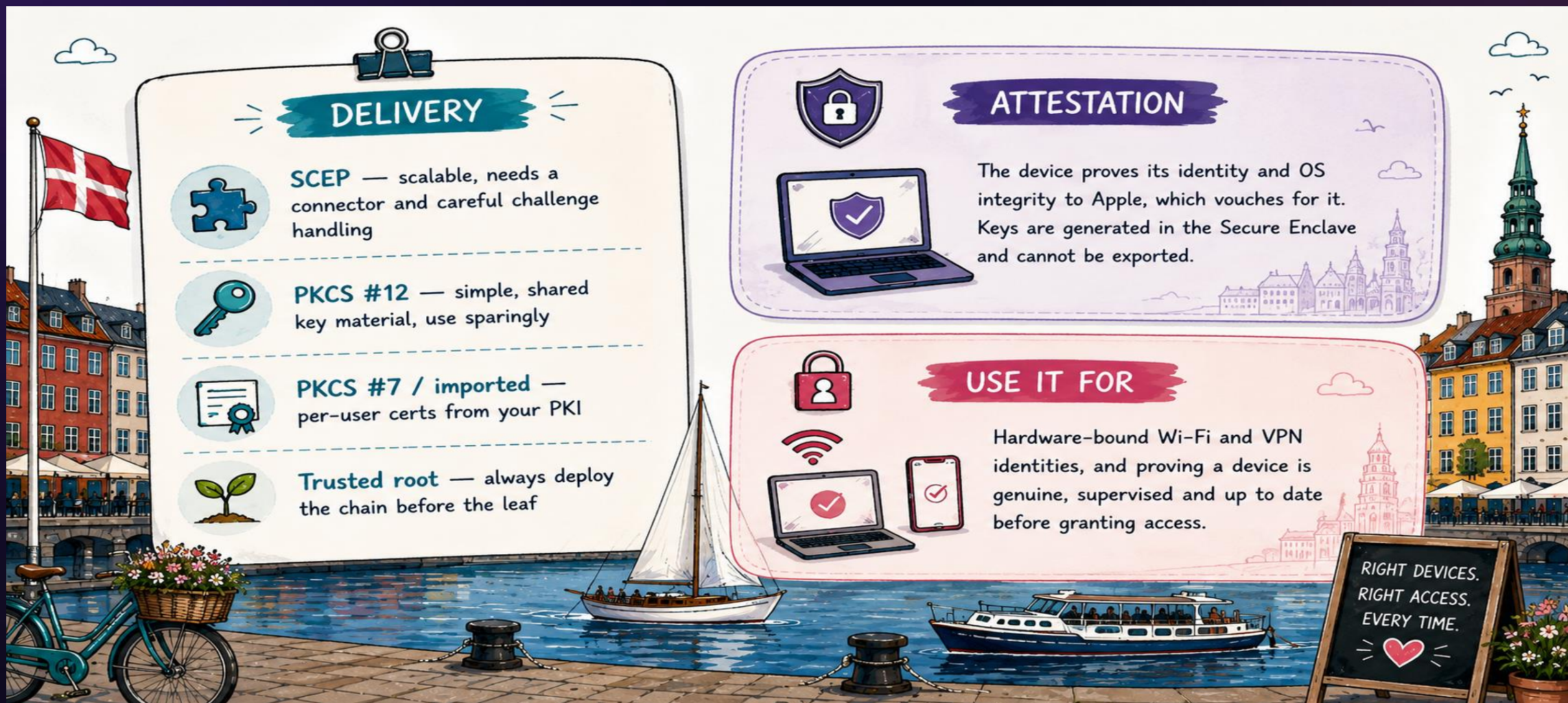
macOS system security — extensions & startup

macOS has unique security layers you must understand and configure before deployment.



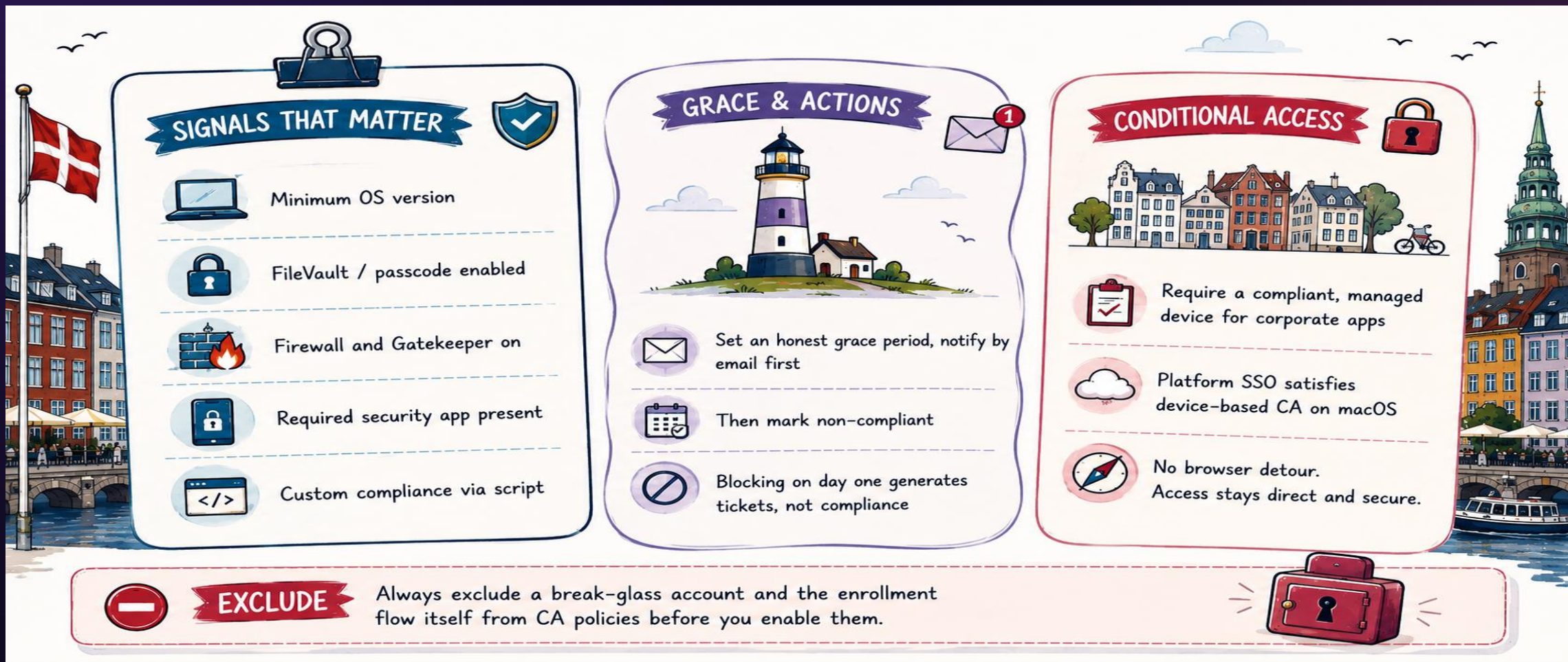
Certificates & Managed Device Attestation

Certificate management and device trust verification are foundational for secure access to corporate resources.



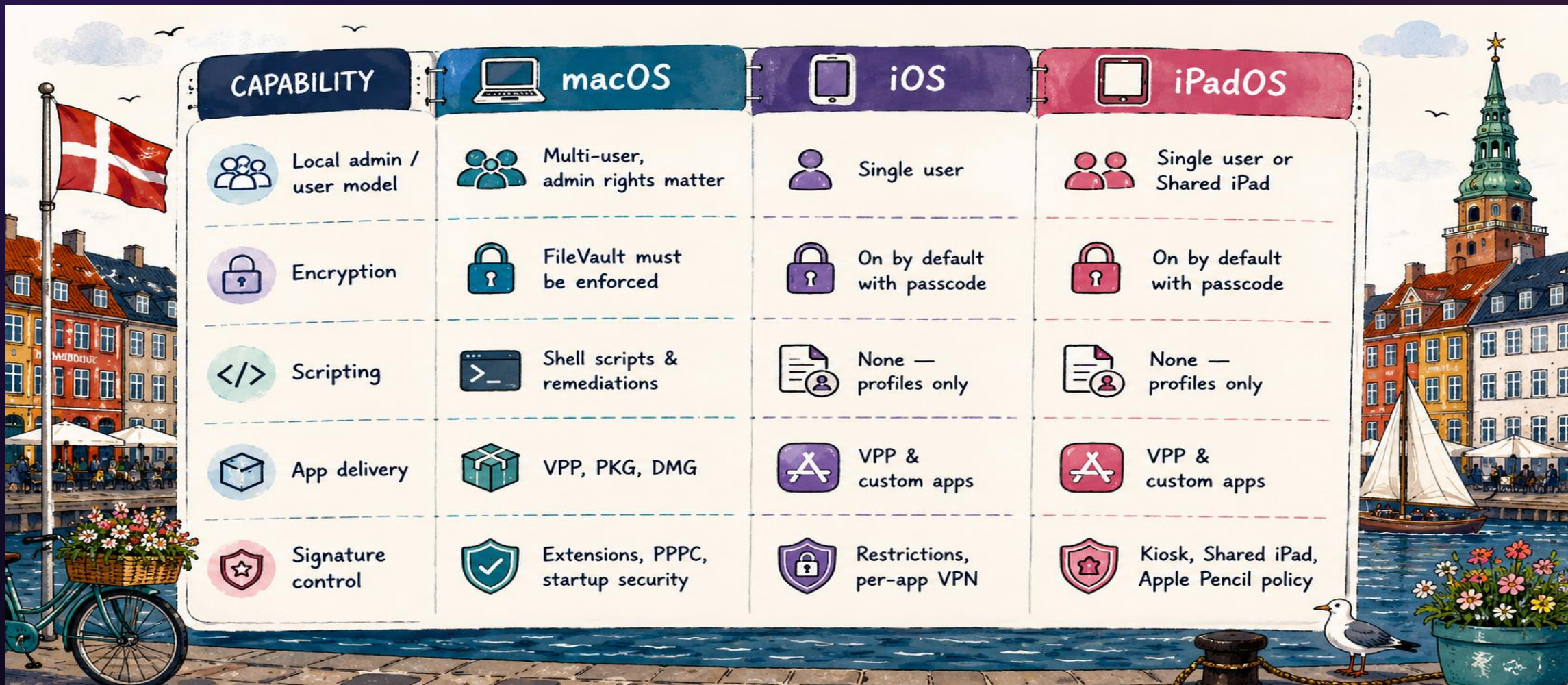
Compliance & Conditional Access

Compliance turns device state into an access decision. Without Conditional Access it is just a dashboard.



macOS vs iOS vs iPadOS

Each platform has unique management capabilities. Plan, configure and test per platform.



CAPABILITY	macOS	iOS	iPadOS
 Local admin / user model	 Multi-user, admin rights matter	 Single user	 Single user or Shared iPad
 Encryption	 FileVault must be enforced	 On by default with passcode	 On by default with passcode
 Scripting	 Shell scripts & remediations	 None — profiles only	 None — profiles only
 App delivery	 VPP, PKG, DMG	 VPP & custom apps	 VPP & custom apps
 Signature control	 Extensions, PPC, startup security	 Restrictions, per-app VPN	 Kiosk, Shared iPad, Apple Pencil policy

Migrating from another MDM

A device can only have one MDM. Migration is unenroll-then-enroll — plan the order, the comms and the fallback.



Troubleshooting & logs

Apple management fails quietly. Knowing where to look turns a two-day ticket into a ten-minute fix.

ON THE MAC

- Log stream — predicate 'subsystem == "com.apple.ManagedClient"'
Live log check
- profiles list -all / profiles status -type enrollment
See profile status
- sudo profiles renew -type enrollment
Force re-enrollment
- System Settings → Device Management
→ profile details
Check profiles in UI
- /Library/Logs/Microsoft/Intune
Gather detailed logs

ON iPhone / iPad

- 1 Install the Apple MDM profiles
logging profile
- 2 Reproduce the issue, then export via Settings
Privacy → Analytics
- 3 Sysdiagnose
for anything Apple support will touch

USUAL SUSPECTS

Expired token	Blocked APNs
Missing bootstrap token	Conflicting profiles
Device not in ABM	Unsupervised device asked to do a supervised thing

Scaling from 1 to 1,000+ devices

Scaling Apple device management requires deliberate planning, automation and clear communication before you grow.

01 GROUPING
Dynamic, not manual

Assign by dynamic groups on enrollment profile, OS and device category. Never hand-add devices.

02 RINGS
Test → pilot → broad

Every profile, app and OS version follows the same route. No exceptions for "small changes".

03 NAMING
Conventions that scale

PROFILE NAME

- ☒ Platform
- ☒ Scope
- ☒ Purpose
- ☒ Version

Platform · scope · purpose · version. Future-you needs to find a profile by name in a list of 300.

04 AUTOMATION
Graph & source control

Export configuration as code, review changes in Git, and re-deploy tenants from a known-good baseline.

05 SELF-SERVICE
Company Portal first

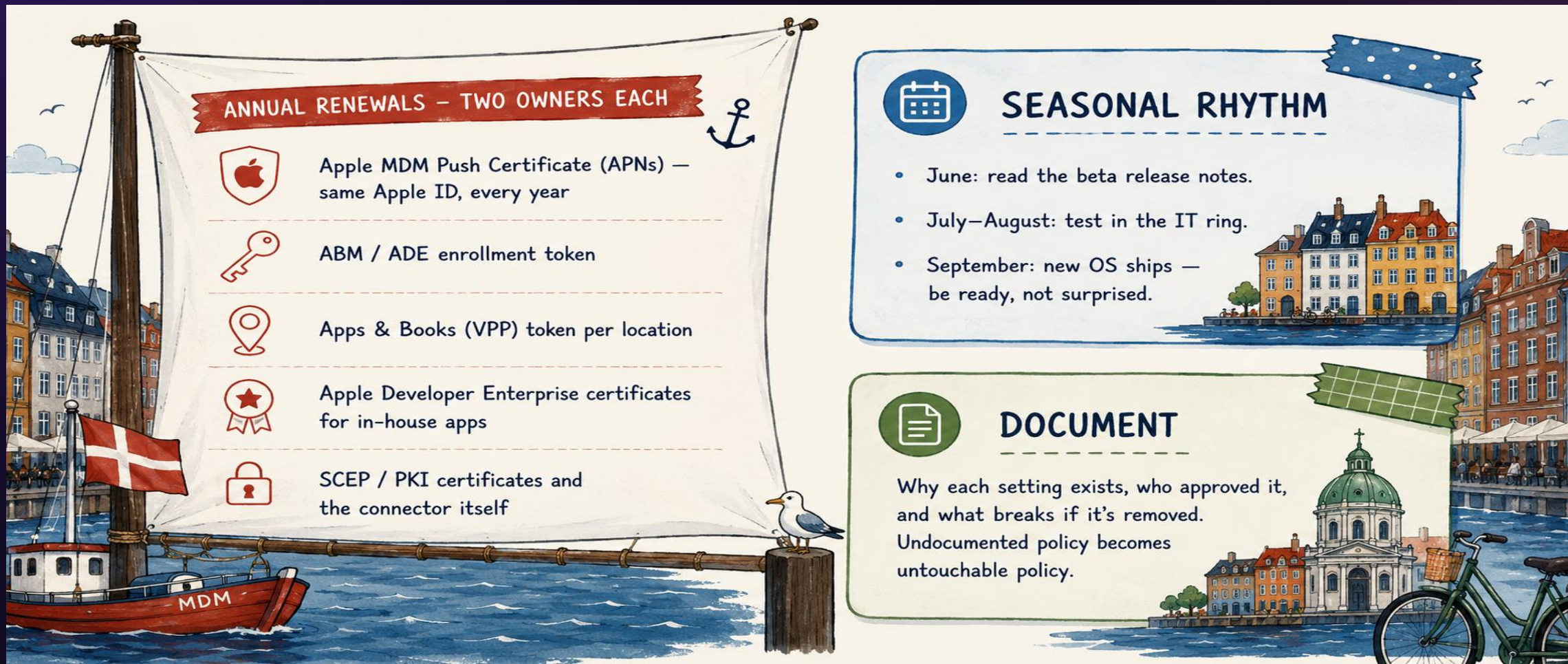
Optional apps, device actions and clear remediation text remove the majority of tickets.

06 PEOPLE
Train the helpdesk

Give first line a runbook: how to check enrollment, retrieve a recovery key, and when to escalate.

Governance — renewals, maintenance & docs

Ongoing governance keeps your deployment healthy and prevents sudden, painful outages.



Gotchas that cost me time



WWDC26

What changes with OS 27

iOS 27, iPadOS 27 and macOS 27 ship this fall. Declarative management stops being optional — and several legacy mechanisms stop working entirely.

What stops working in OS 27

These aren't deprecations with a grace period. They fail — and mostly they fail quietly.



LEGACY UPDATE

Legacy update MDM removed

Update commands, queries, recommended counters, deferred and background actions. Improvements maintained at step-tracking. ECM software updates are the only path.



SECURITY STANDARDS

TLS 1.2 minimum, ATS-compliant

Enforced, profile-tested, app-held and updates all require it. Exceptions are short-term — track with approval and expire on 12/2026 before September.



PROFILE CONTROL

Management no longer restores

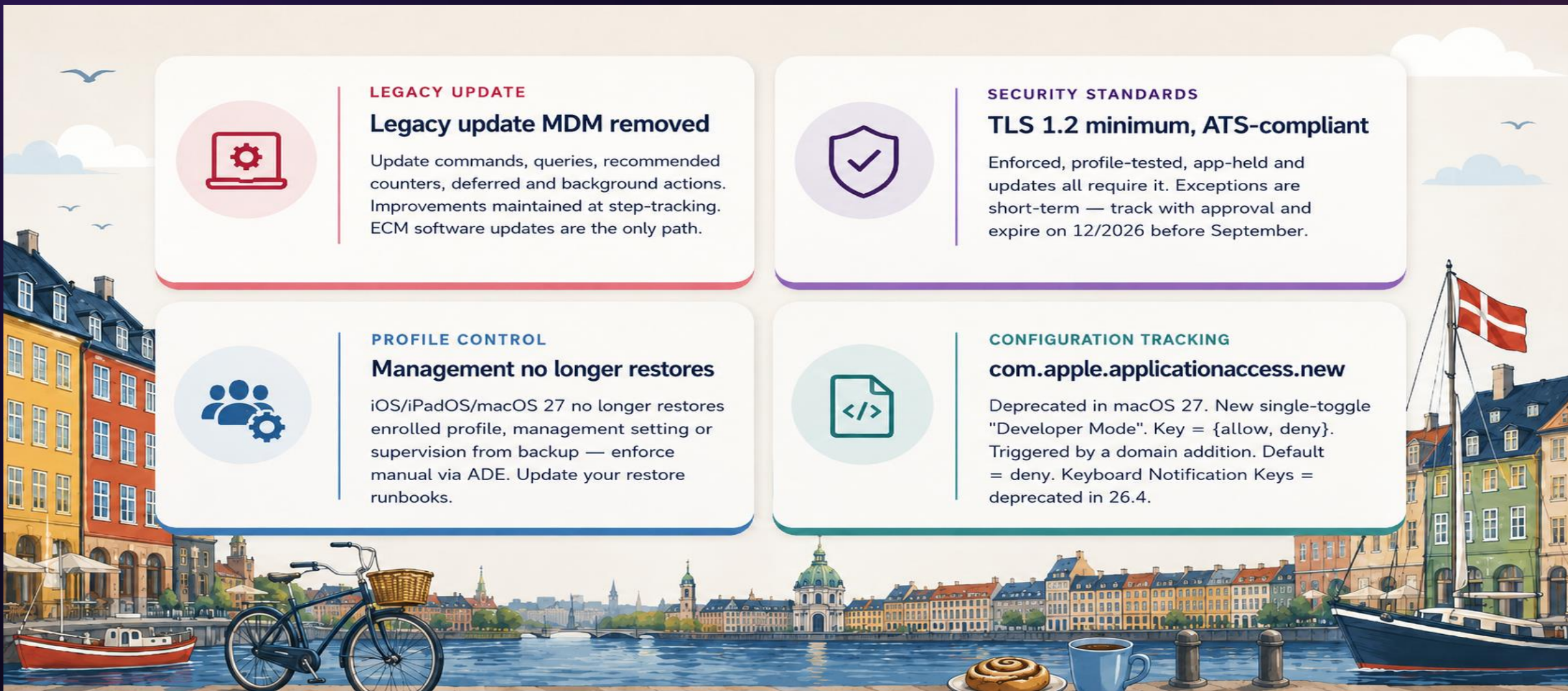
iOS/iPadOS/macOS 27 no longer restores enrolled profile, management setting or supervision from backup — enforce manual via ADE. Update your restore runbooks.



CONFIGURATION TRACKING

`com.apple.applicationaccess.new`

Deprecated in macOS 27. New single-toggle "Developer Mode". Key = {allow, deny}. Triggered by a domain addition. Default = deny. Keyboard Notification Keys = deprecated in 26.4.



New declarative configurations

Apple keeps expanding what a declaration can express — and is explicitly asking admins and vendors to build on DDM first.



DECLARATIVE VPN

IKEv2, IPSec, Always-On, DNS proxy, DNS configs: ready for easy decisions.



CREDENTIALS

Declarative credential management. Certificates do what you declare — no surprises.



APPS

Unified app settings. One place for allowlists, certs, and proxies. Less hopping, more productivity.



PRIVACY

Consolidated consent. Privacy keys in one view with clear controls. PPE/TCC: simplified.



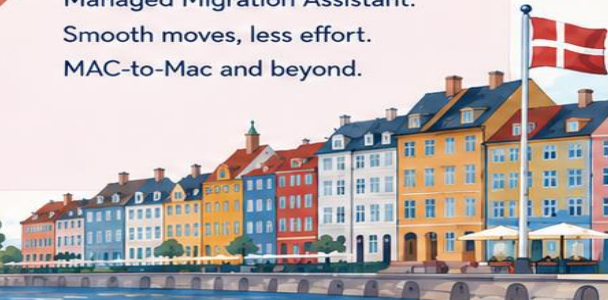
WEB FILTER

Content caching & web filter. Faster browsing, secure access with smart configurations.



MIGRATION

Managed Migration Assistant. Smooth moves, less effort. MAC-to-Mac and beyond.



Platform SSO grows up

Extensible SSO and Platform SSO both move into the declarative model, and macOS 27 adds web-based authentication at the screens that mattered most.



LOGIN & ACCESS

- IdP web-view of the FW lock screen
- IdP web-view of the lock screen and login window
- QR code sign-in
- Offline grace period for disconnected users
- Declarative: `com.apple.access/accessible-azo`



PAY IT MATTERS

Conditional Access and MFA match the preferences screens, and the kiosk-to-Entra password age that has conf, local users since POSD skipped 1-day close.



ALSO PLAN FOR

Microsoft ending & debbling: Exchange Web Services on 1 Oct 2026, permanently off 1 Apr 2027. Apple's Graph API support for Mail, Calendar, Contacts, Notes and Reminders arrives in all later macOS 27 updates — not all OS.

Managing Apple Intelligence & Siri

AI feature management moves into DDM. The legacy restriction keys were deprecated in the 26.4 releases — the runway is ending.



NEW CONFIGURATIONS & KEYS

- `com.apple.configuration.intelligence.settings`
- `com.apple.configuration.external-intelligence.settings`
- `com.apple.configuration.siri.settings`
- `com.apple.configuration.keyboard.settings`



AllowGameobj



AllowImagePlayground



AllowWritingTools



AllowImageAnd
per-app AI
controls



SET EXPECTATIONS

The new Siri launches in English with a waitlist is unavailable in the EU on iPhone and iPad, and the most capable on-device model needs 12 GB of unified memory — else it falls back to Private Cloud Compute.



DO THIS

Review each control against your acceptable-use policy before OS 27, and decide per-ring rather than banning everything by default.

New status items & device health

Status subscriptions get real teeth: compliance checks that don't depend on a proprietary agent.



DEVICE INTEGRITY

Hardware genuineness

Reports whether baseband, camera, Face ID, Touch ID, NFC and UWB components are genuine — tamper detection on iPhone and iPad.



ENROLLMENT STATE

Granular MDM status

- ✓ `mdm.enrollment-type`
- ✓ `mdm.is-awaiting-configuration`
- ✓ `mdm.is-return-to-service`
- ✓ `mdm.is-shared-ipad`
- ✓ `security.lockdown-mode`



SUPPORT

Remote log collection

New commands trigger and cancel enhanced log collection on supervised devices for AppleCare cases — an AppleCare Enterprise agreement applies.



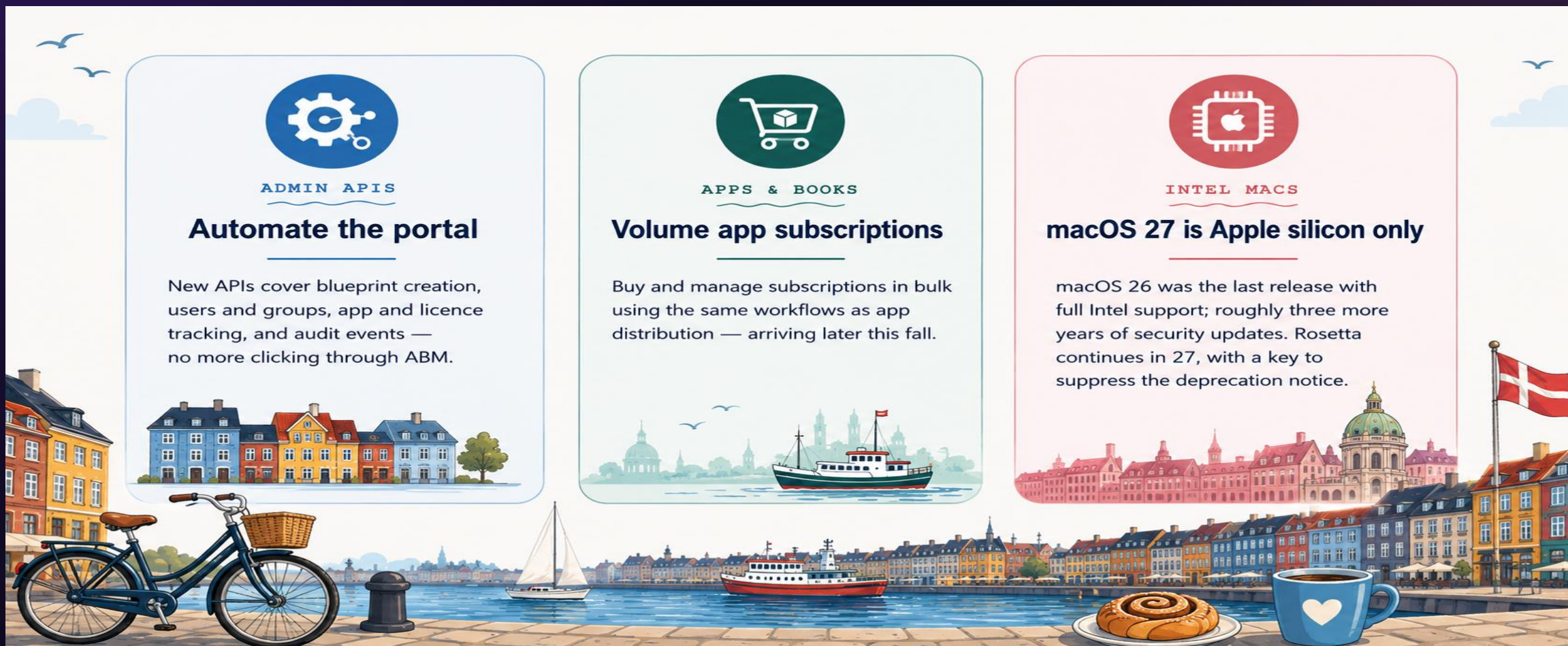
RETURN TO SERVICE

New keys retry enrollment automatically with backoff up to **5 minutes**, and set language and region — **fewer stuck devices** in the wipe-and-reissue loop.



Apple Business platform & app management

Zero-touch deployment, Managed Apple Accounts and built-in device management now sit on one platform, available in 200+ countries and regions.





ADMIN APIS

Automate the portal

New APIs cover blueprint creation, users and groups, app and licence tracking, and audit events — no more clicking through ABM.



APPS & BOOKS

Volume app subscriptions

Buy and manage subscriptions in bulk using the same workflows as app distribution — arriving later this fall.



INTEL MACS

macOS 27 is Apple silicon only

macOS 26 was the last release with full Intel support; roughly three more years of security updates. Rosetta continues in 27, with a key to suppress the deprecation notice.

Your OS 27 action plan



MDM Priorities – On Deck

01



Confirm DDM software update is shipped.

Today, not in September.



02



Audit TLS/ATS compliance

across MDM and all internal management endpoints.



03



Migrate Intelligence, Siri & keyboard restrictions
to the new declarations.



04



Plan the move off

`com.apple.applicationaccess.new`
and the TCC profile keys.



05



Rewrite restore & re-enrollment runbooks
for the backup change.



06



Start the Intel refresh conversation
and pilot Managed Migration Assistant.



Pre-release detail as announced at WWDC26 (8–12 June 2026);
verify keys against Apple's deployment documentation as betas progress.



Key takeaways & resources

01



Get ABM right first —
identity and enrollment are the foundation everything else sits on.

02



Supervise through ADE.
It is a **purchase-time** and **wipe-time** decision, not a setting.

03



Platform SSO plus Conditional Access
gives you security users don't fight.

04



Move to declarative management
and enforced update deadlines.

05



Automate, document, and diary every renewal.
Governance is the whole job at scale.



GO DEEPER



Apple Platform Deployment & Security guides



support.apple.com —
Apple Business Manager User Guide



learn.microsoft.com —
Intune for macOS & iOS



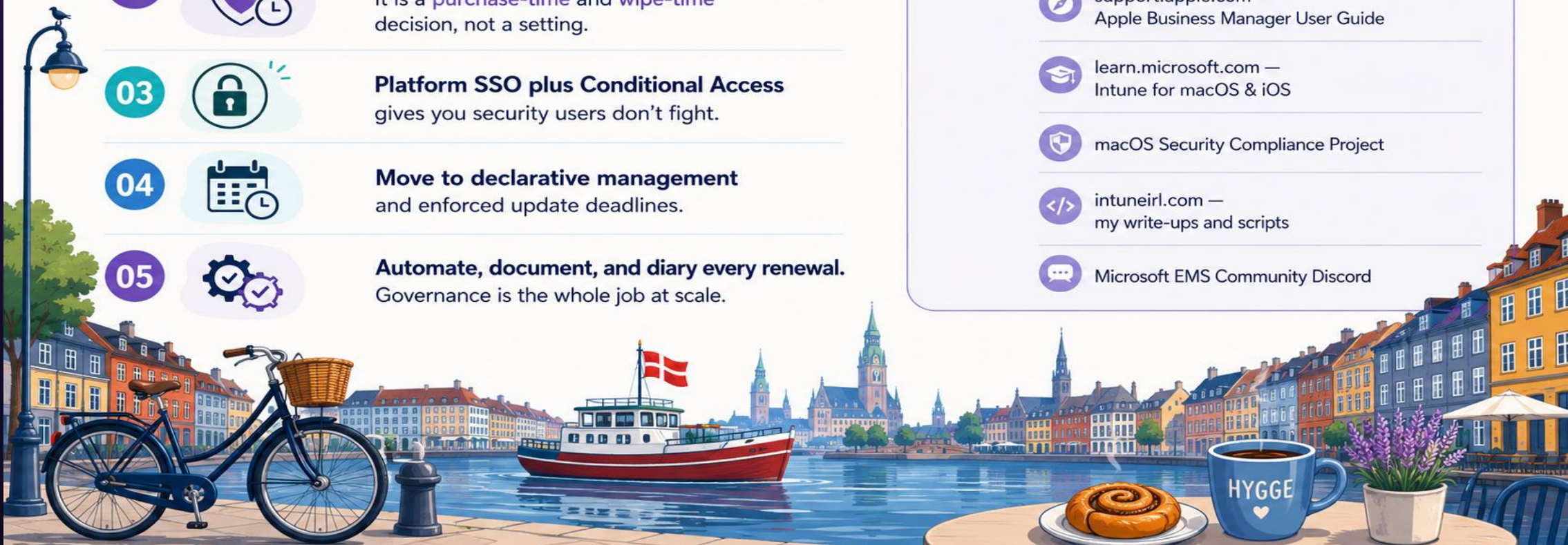
macOS Security Compliance Project



intuneirl.com —
my write-ups and scripts



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